

GROUP 'B' (5x5=25)

SHORT ANSWER TYPE QUESTIONS.

10. Why we use multimedia? Define linear and non-linear multimedia with examples. Explain the different components of multimedia. [1+1+3]
11. Explain the memory hierarchy in a computer system with a neat and labeled diagram. Describe the different levels of memory and discuss their characteristics in terms of speed, cost, and capacity. [2.5+2.5]
12. Describe any three types of positional number system. Convert the following as indicated. [3+1+1]
- a) $(93.6)_{10} = (?)_2$ b) $(BAD.DAD)_{16} = (?)_8$

'OR'

Subtract 1011_2 from 10101_2 using 1's and 2's complement methods. [2.5+2.5]

13. What is mobile computing? Suppose the internet and mobile networks suddenly stopped working for one full week. Describe at least two problems people would face in education, business, and daily life. Suggest at least two alternative solutions people could use during that time. [1+1+1+1+1]

'OR'

Explain in detail the concept of mail merge in Microsoft Word. What are its primary benefits and disadvantages? [5]

14. Define output devices with examples. Write their functions. Write any four differences between CRT monitor and LCD monitor. [1+2+2]

GROUP 'C' [2×8=16]

LONG ANSWER TYPE QUESTIONS.

15. What do you mean by computer architecture? Describe each functional components of computer architecture with typical diagram. [1+7]
16. You are given to prepare a project report in MS Word about "Generations of Computers."
- a) Explain how you would use any four Word features (like Table of Contents, Page Numbering, WordArt, Styles, SmartArt, etc.) to organize the report neatly. [4×1.5=6]
- b) Write why these features are useful for readers. [2]

"Best of Luck"

State and use parallelogram law of vector addition to find the magnitude and direction of the resultant of two vectors inclined at an angle θ .
 Two forces F_1 and F_2 act at a point. F_1 has magnitude 8N and acts at an angle of 30° with the positive x-axis. F_2 has a magnitude of 6N and acts at an angle of 150° with positive x-axis. Find the magnitude and direction of resultant force.
 Derive a relation between focal length and radius of curvature of a spherical mirror.
 A small candle 2.5 cm in size is placed 27 cm in front of a concave mirror of radius of curvature 36 cm. At what distance from the mirror should a screen be placed in order to receive a sharp image? Describe the nature & size of the image.
 Derive a mirror formula for concave mirror and mention the sign convention.
 A concave mirror forms an image 3 times as large as an object placed 0.3 m from it. Calculate its focal length.
 Define the coefficient linear, superficial and cubical expansions. Derive relations between them.
 Why a thick glass cracks when boiling water is poured in it?
 Pendulum clocks generally go fast in winter and slow in summer. Why?
 A pendulum clock loses 12 Sec. a day if the temperature is 40°C and gains 4 Sec. a day if the temperature is 20°C . Find the temperature at which the clock will show correct time and coefficient of linear expansion of the metal of pendulum shaft.
 State and explain the principle of superposition of Coulomb's force.
 Two point charges of $10\ \mu\text{C}$ and $-40\ \mu\text{C}$ are separated by distance of 13 cm. Find the point along the line joining them at which net electrostatic force on any charge is zero.
 Define dielectric constant. Why its value is always greater than one?
 Four equal charges q are placed at the four corners of a square of length a . Find the magnitude of the net force acting on one of them due to other charges and draw diagram to show the direction of force on that charge.

GROUP 'C' [3×8=24]

Answer questions:
 Explain why vector addition is not commutative, providing a real-life example to support your reasoning.
 What is meant by resolution of vectors? How will you resolve a vector in a plane in two mutually perpendicular directions?
 A plane flies 200 km at 60° north of east, then 150 km at 30° south of west and finally 100km at 45° south of east. Calculate the magnitude and direction of the resultant displacement.
 Why does a solid expand on heating?
 Design an experiment to determine the coefficient of linear expansion of a solid.
 A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder at 15°C is 10 cm and there is an all-round clearance of 0.05 mm between the piston and the cylinder of the wall. Find the temperature at which the two will exactly fit in.
 $\alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$ and $\alpha_{\text{aluminium}} = 1.6 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$
 State and explain Coulomb's law in electrostatics.
 Coulomb's law is applicable to stationary charges only?
 Calculate the force between two 1 coulomb charge on the basis of Coulomb's law.
 Two charges 10^{-6} C and 10^{-7} C attract each other with a force of 400N. Calculate the distance between them if they are in
 i) air and
 ii) water (relative permittivity of water is 80)

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Best of Luck